



UNIVERSITY OF PUNJAB

Faculty of Computing & Information Technology

COURSE CODE: CC-215-L | DATABASE SYSTEMS LAB

LAB 01

SQL — SELECT Statement & WHERE Clause

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Total Marks: 100

1. Instructions

- Use the same database you created for Pre-Lab 01 (setup-db.sql). If you have not run it yet, do so before starting Task 1.
- Compile all screenshots into a single Word document, clearly labeled Q1 through Q20, in the order given below.
- Name the file using your Roll Number followed by Lab01 (e.g., RollNo_Lab01.docx).

Task 1 — Basic SELECT & WHERE (Easy) (10 questions × 4 marks = 40 marks)

- Q1.** Display all columns and all rows from the Users table.
- Q2.** Display the name and country of all users who joined after '2021-01-01'.
- Q3.** Display the distinct countries present in the Users table.
- Q4.** Display all channels whose subscriber_count is between 100000 and 300000 (inclusive).
- Q5.** Display all videos whose category is either 'Technology' or 'Gaming'.
- Q6.** Display all users whose name starts with the letter 'A'.
- Q7.** Display all comments with a rating of 5 and more than 300 likes.
- Q8.** Display the title, category, and ad_revenue of all videos, sorted by ad_revenue from highest to lowest.
- Q9.** Display the title, views, and duration_seconds of videos that are shorter than 900 seconds, sorted by views in descending order.
- Q10.** Display the channel_name and subscriber_count of the top 5 channels with the highest subscriber_count.

Task 2 — Operators, Filtering & Sorting (Medium) (7 questions × 5 marks = 35 marks)

Q11. Display all videos where the category is NOT 'Gaming' and the views are greater than 50000 (use the NOT and AND logical operators).

Q12. Display the channel_name and subscriber_count of channels whose subscriber_count is greater than 200000 AND whose channel_name starts with either 'T' or 'S' (combine AND, OR, and LIKE, using parentheses to group the OR condition).

Q13. Display the title, views, and ad_revenue of every video, along with a computed column revenue_per_1000_views calculated as ad_revenue divided by (views / 1000), using an arithmetic operator in the SELECT list.

Q14. Display all comments where the rating is between 3 and 5 OR the number of likes is greater than 500, sorted by likes in descending order.

Q15. Display the name and joined_date of users who joined between '2020-01-01' and '2022-12-31' (inclusive), sorted by joined_date in ascending order.

Q16. Display the title and duration_seconds of videos whose duration_seconds is NOT between 300 and 1200, sorted by duration_seconds in ascending order.

Q17. Display the channel_name and subscriber_count of channels ranked 6th to 10th by subscriber_count (use ORDER BY together with LIMIT and OFFSET).

Task 3 — Combined Conditions & Ranking (Difficult) (3 questions = 25 marks)

Q18. Display the title, category, views, and ad_revenue of videos where (category = 'Technology' AND views > 100000) OR (category = 'Gaming' AND ad_revenue > 5000), sorted by ad_revenue from highest to lowest. (Combine AND, OR, and parentheses to control operator precedence.)

Q19. Display the name, country, and joined_date of users who are NOT from 'USA' and NOT from 'UK', and whose name does NOT start with the letter 'A' or 'E', sorted alphabetically by name. (Combine multiple NOT/AND conditions with LIKE.)

Q20. Display the video_id, title, views, and duration_seconds of the 3 videos with the highest views among those where duration_seconds is greater than 600 AND (views > 1000000 OR ad_revenue > 10000). (Combine WHERE with AND/OR, ORDER BY, and LIMIT.)

2. Submission Instructions

- Solve all 20 questions above using the database from Pre-Lab 01's setup script.
- Create a new Word document and paste each question's query and screenshot into it, clearly labeled Q1 through Q20, in order, grouped under Task 1, Task 2, and Task 3 headings.
- Name the file using your Roll Number followed by Lab01 (e.g., RollNo_Lab01.docx).
- This lab is graded and carries 100 marks, distributed as shown above.